Candidate’s Name: Azam Azri Ahmad

1. Are you currently employed?

I am currently pursuing a Master's degree in Information Technology. I have previously completed multiple freelance web development projects, including e-commerce and business websites. My academic commitments are well-managed, allowing me to focus fully on a remote full-time role.

2. What motivated you to become a web developer?

My motivation stems from the unique combination of creativity and problem-solving that web development offers. I'm fascinated by how code can transform ideas into functional, user-facing applications that solve real business problems. Through my academic research in AI systems and hands-on commercial projects, I've developed a passion for building scalable, efficient web solutions that deliver tangible value to users and businesses.

3. Can you explain the difference between HTML, CSS, and JavaScript?

Hypertext Markup Language (HTML) is the structural foundation of web pages, defining content elements like headings, paragraphs, and links. Cascading Style Sheets (CSS) controls the visual presentation and layout, including colors, typography, spacing, and responsive design. JavaScript is the programming language that adds interactivity and dynamic behavior, enabling features like form validation, animations, and real-time content updates. Together, they form the core trinity of front-end web development: HTML provides structure, CSS handles aesthetics, and JavaScript delivers functionality.

4. How do you ensure your code is accessible to users with disabilities?

I implement Web Content Accessibility Guidelines (WCAG) standards by using semantic HTML elements, proper heading hierarchy, and ARIA (Accessible Rich Internet Applications) labels for dynamic content. I ensure sufficient color contrast ratios, provide alternative text for images, enable keyboard navigation for all interactive elements, and test with screen readers. In my recent projects using Next.js and TypeScript, I incorporate accessibility checks into my development workflow and use tools like Lighthouse to audit accessibility scores before deployment.

5. Describe the difference between GET and POST requests.

GET requests retrieve data from a server and append parameters to the Uniform Resource Locator (URL), making them visible and bookmarkable. They're idempotent, meaning multiple identical requests produce the same result without modifying server state. POST requests send data in the request body, suitable for submitting forms or creating resources.

They can handle larger payloads, keep sensitive data hidden from URLs, and are used for operations that modify server state. GET is appropriate for searches and filtering, while POST is essential for user authentication, data creation, and file uploads.

6. How would you optimize a website's performance?

I employ multiple optimization strategies: implementing code splitting and lazy loading to reduce initial bundle size, optimizing images through modern formats like WebP and appropriate compression, leveraging Content Delivery Networks (CDNs) for static assets, and implementing server-side rendering or static site generation with Next.js. I minimize JavaScript execution time, eliminate render-blocking resources, use efficient caching strategies, and optimize database queries. In my AWS-based projects, I utilize cloud services for scalability and implement performance monitoring to identify bottlenecks continuously.

7. How do you handle cross-browser compatibility issues?

I use feature detection rather than browser detection, implement progressive enhancement strategies, and test across major browsers during development. I leverage CSS vendor prefixes through tools like Autoprefixer, use polyfills for newer JavaScript features when targeting older browsers, and validate code against web standards. My TypeScript and Tailwind CSS workflow helps maintain consistency, and I utilize BrowserStack or similar tools for comprehensive cross-browser testing before deployment.

8. What are the differences between server-side and client-side programming?

Server-side programming executes on the web server, handling business logic, database operations, authentication, and generating dynamic content before sending it to the client. It provides better security for sensitive operations and Search Engine Optimization (SEO) benefits. Client-side programming runs in the user's browser, enabling immediate user interactions, reducing server load, and providing responsive interfaces without page reloads. My experience with Next.js allows me to leverage both approaches strategically—using server-side rendering for initial page loads and SEO-critical content, while client-side JavaScript handles interactive features.

9. How do you secure a web application against common vulnerabilities?

I implement multiple security layers: validating and sanitizing all user inputs to prevent Cross-Site Scripting (XSS) and Structured Query Language (SQL) injection attacks, using parameterized queries with Supabase, implementing Cross-Site Request Forgery (CSRF) tokens, enforcing HTTPS, and setting appropriate security headers. I follow the principle of least privilege for database access, implement proper authentication and authorization using secure session management, regularly update dependencies to patch vulnerabilities,

and never expose sensitive credentials in client-side code. My AWS deployments include security groups and proper Identity and Access Management (IAM) configurations.

10. How would you approach collaborating with designers and other developers on a project?

I prioritize clear communication and establish shared workflows from the start. I work with designers to understand design systems, ensure technical feasibility, and maintain consistency through component libraries. I use version control with Git for collaborative development, write clear documentation, conduct code reviews, and follow established coding standards. In my recent projects, I've successfully coordinated with teams using project management tools, maintained clear commit messages, and created reusable components that bridge design and development efficiently.

11. Can you explain the concept of asynchronous programming?

Asynchronous programming allows code execution to continue without waiting for time-consuming operations to complete, preventing application blocking. In JavaScript, this is managed through callbacks, Promises, and async/await syntax. When fetching data from Application Programming Interfaces (APIs), asynchronous operations enable the interface to remain responsive while waiting for server responses. I regularly implement asynchronous patterns in my TypeScript projects for API calls, file uploads, and database operations, ensuring optimal user experience and application performance.

12. Tell me about your experience working remotely.

I have extensive remote work experience managing multiple concurrent projects as a freelance developer while pursuing graduate studies. I maintain disciplined work schedules, use project management tools for organization, and communicate proactively with clients across different time zones. My remote workflow includes using cloud-based development environments, version control systems, and collaboration platforms. I'm self-motivated, meet deadlines consistently, and have successfully delivered complex projects including e-commerce platforms and business websites entirely through remote collaboration.

13. Can you discuss a recent web development project you worked on, including challenges you faced and how you overcame them?

I recently worked on developing a business website for a service-based company that required dynamic content management, strong SEO performance, and a smooth user experience across devices. The website included multiple service pages, media-rich sections, and an admin interface for updating content without technical assistance.

One of the main challenges was maintaining fast load times and SEO performance while handling dynamic content and high-resolution images. I addressed this by optimizing images, implementing efficient data-fetching strategies, and applying server-side rendering for SEO-critical pages. Another challenge was ensuring stable deployment and secure configuration. I resolved this by restructuring API logic, properly managing environment variables, and refining the deployment setup, resulting in a stable, scalable, and well-performing production website.

14. Describe your daily routine as a web developer.

My day begins with reviewing project priorities and checking communications from clients or collaborators. I dedicate focused blocks to coding, typically starting with the most complex problems when my concentration is highest. I regularly commit code with clear messages, conduct self-reviews, and test features across different scenarios. I allocate time for learning new technologies and staying current with web development trends through documentation and technical articles. I maintain detailed documentation as I work, conduct performance testing, and end my day by planning the next day's tasks and updating project progress.

15. What makes you stand out from other web developer candidates for this positi

My unique combination of advanced technical expertise and academic research background sets me apart. I bring production-ready experience with modern technologies like Next.js, TypeScript, and cloud architecture while applying analytical rigor from my Master's research in AI and machine learning. I've successfully delivered commercial projects independently, demonstrating self-management skills crucial for remote work. My experience with advanced system architecture, database design, and full-stack development enables me to contribute immediately to complex projects. Additionally, my ongoing research in conversational AI and natural language processing provides innovative perspectives on user experience and intelligent web applications that can benefit forward-thinking projects.